

Assignment No 3

1. Problem 1: Sum of Two Numbers (Using a Method)

1.1 Code

```
import java.util.Scanner;

public class SumTwoNum
{
    static int sumOfTwoNumbers(int a, int b)
    {
        return a+b;
    }
    public static void main(String args[])
    {
        Scanner sc = new Scanner(System.in);

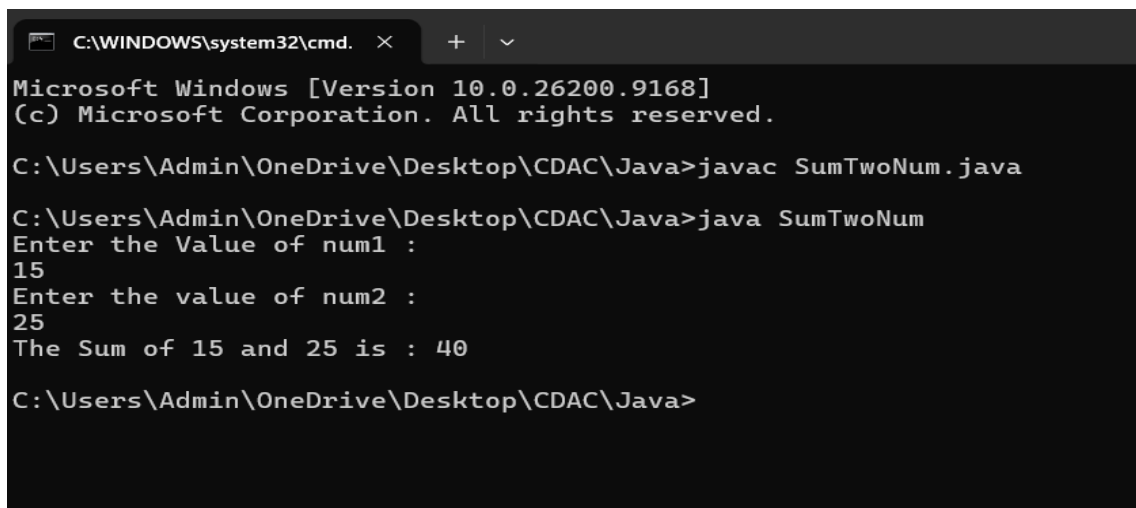
        System.out.println("Enter the Value of num1 : ");
        int num1 = sc.nextInt();

        System.out.println("Enter the value of num2 : ");
        int num2 = sc.nextInt();

        int sum = sumOfTwoNumbers(num1, num2);

        System.out.println("The Sum of " + num1 + " and " + num2 +
        " is : " + sum);
    }
}
```

1.2 Output:



```
C:\WINDOWS\system32\cmd.  x  +  v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac SumTwoNum.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java SumTwoNum
Enter the Value of num1 :
15
Enter the value of num2 :
25
The Sum of 15 and 25 is : 40

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

2. Simple Age Checker (Using a Method)

2.1 Code

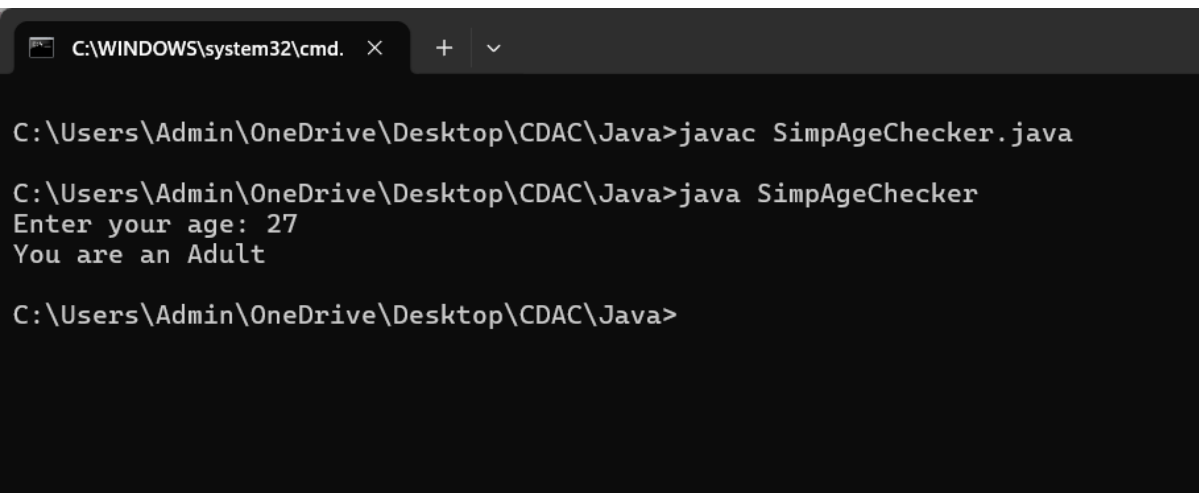
```
import java.util.Scanner;

public class SimpAgeChecker
{
    static void checkAgeCategory(int age)
    {
        String data = (age < 18) ? " You are a Minor " : (age < 60) ? "You are
an Adult" : "You are a senior citizen";
        System.out.println(data);
    }
    public static void main(String args[])
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        checkAgeCategory(age);
    }
}
```

2.2 Output



```
C:\WINDOWS\system32\cmd. x + v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac SimpAgeChecker.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java SimpAgeChecker
Enter your age: 27
You are an Adult

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

3. Problem 3: Print Even Numbers (Using while Loop)

3.1 Code

```
public class EvenNumUsingWhileLoop
{
    static void printEvenNumbers()
    {
        int n = 1;
        System.out.println("Even Numbers are : ");

        while(n <= 50)
        {
            if( n % 2 == 0)
            {

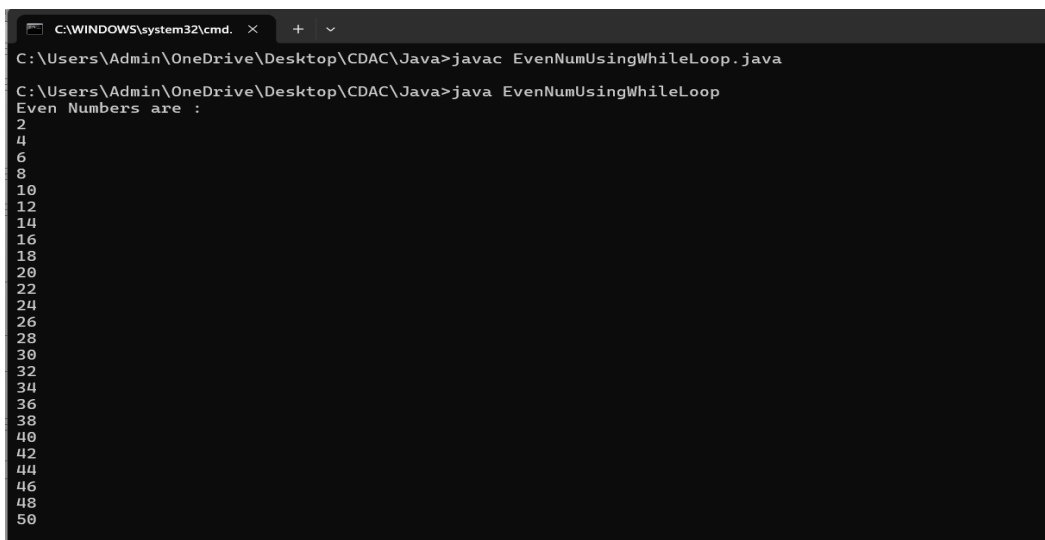
                System.out.println(n);

            }
            n++;
        }

    }

    public static void main(String args[])
    {
        printEvenNumbers();
    }
}
```

3.2 Output



```
C:\WINDOWS\system32\cmd. x + v
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac EvenNumUsingWhileLoop.java
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java EvenNumUsingWhileLoop
Even Numbers are :
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
```

4. Problem 4: User Input for Positive Numbers (Using do-while Loop)

4.1 Code

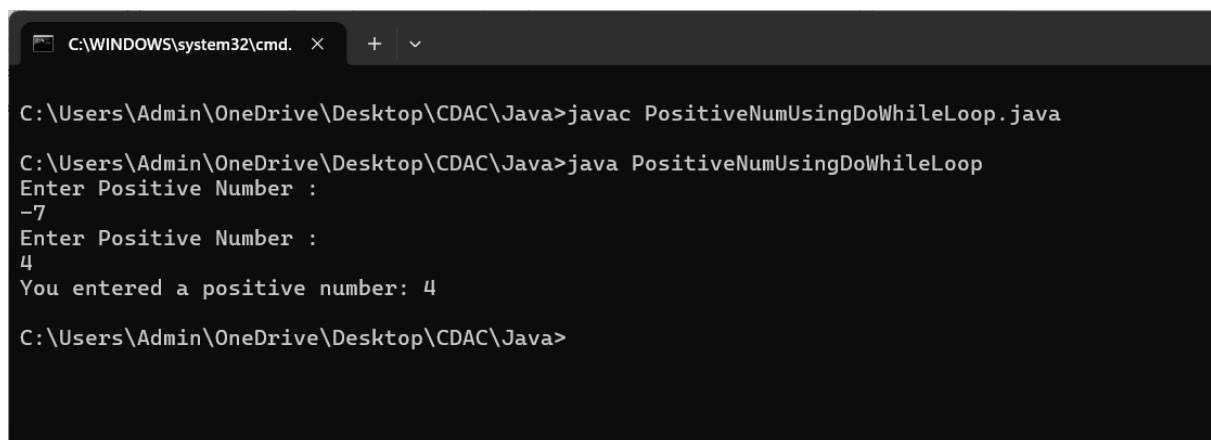
```
import java.util.Scanner;

public class PositiveNumUsingDoWhileLoop
{
    static void askForPositiveNumber()
    {
        Scanner obj = new Scanner(System.in);
        int n;

        do
        {
            System.out.println("Enter Positive Number : ");
            n = obj.nextInt();
        }
        while(n <= 0);
        {
            System.out.println("You entered a positive number: " + n);
        }
    }

    public static void main(String args[])
    {
        askForPositiveNumber();
    }
}
```

4.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac PositiveNumUsingDoWhileLoop.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java PositiveNumUsingDoWhileLoop
Enter Positive Number :
-7
Enter Positive Number :
4
You entered a positive number: 4

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

5. Problem 5: Print Multiplication Table (Using for Loop)

5.1 Code

```
import java.util.Scanner;

public class MultiplicationTableUsingForLoop
{

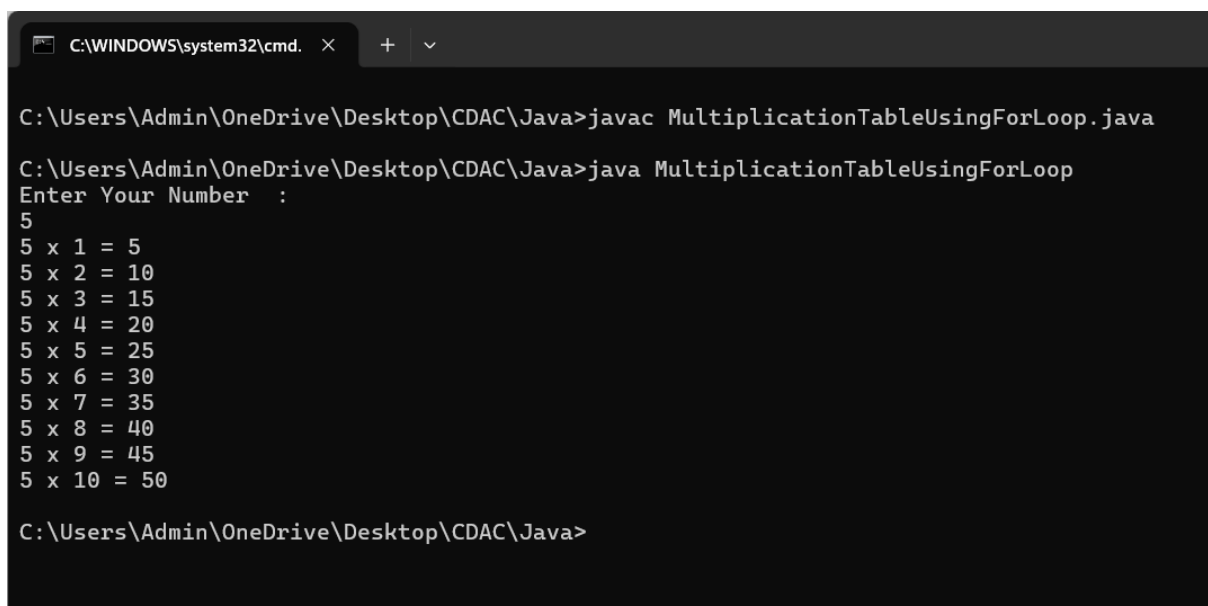
    static void printMultiplicationTable(int number)
    {
        for (int i = 1; i <= 10; i++)
        {
            System.out.println(number + " x " + i + " = " + (number * i));
        }
    }

    public static void main(String args[])
    {
        Scanner obj = new Scanner(System.in);

        System.out.println("Enter Your Number :");
        int num = obj.nextInt();

        printMultiplicationTable(num);
    }
}
```

5.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac MultiplicationTableUsingForLoop.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java MultiplicationTableUsingForLoop
Enter Your Number :
5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

6. Problem 6: Calculate the Sum of Numbers from 1 to N (Using for Loop)

6.1 Code

```
import java.util.Scanner;

public class SumOfNNumUsingForLoop
{
    static void calculateSum(int n)
    {
        int sum = 0;

        for (int i = 1; i <= n; i++)
        {
            sum = sum + i;
        }

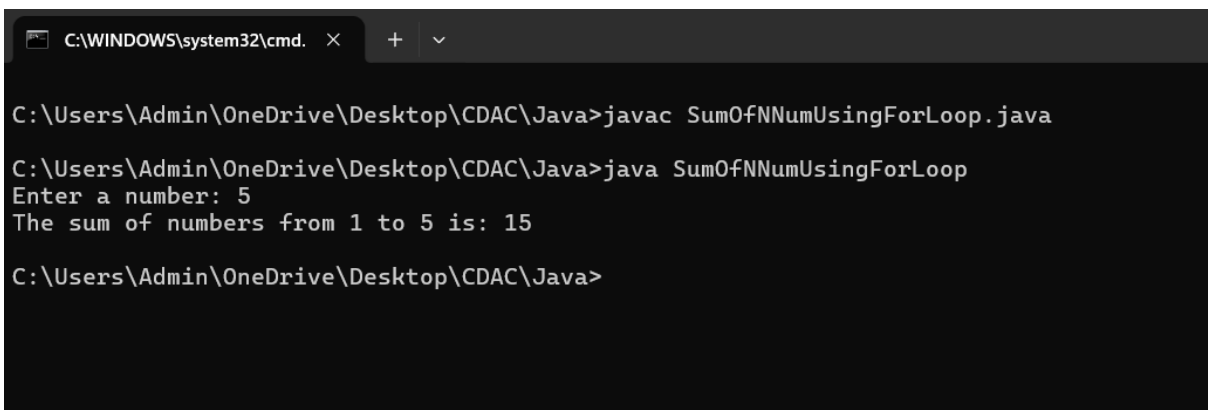
        System.out.println("The sum of numbers from 1 to " + n + " is: " +
sum);
    }

    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        calculateSum(num);
    }
}
```

6.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac SumOfNNumUsingForLoop.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java SumOfNNumUsingForLoop
Enter a number: 5
The sum of numbers from 1 to 5 is: 15

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

Bonus Problem: Menu-Driven Java Program (Switch-Case)

Code:

```
import java.util.Scanner;

public class MenuDrivenUsingSwitchCase
{
    public static void main(String args[])
    {
        Scanner obj = new Scanner(System.in);

        int choice;

        do
        {
            System.out.println("\nMenu-Driven are : ");
            System.out.println("1.Grade Evaluation System");
            System.out.println("2.Leap Year Check");
            System.out.println("3.Day of the week ");
            System.out.println("4.Identify Default Values of Variables ");
            System.out.println("5.Exit\n");

            System.out.println("\nEnter Your choice : ");
            choice = obj.nextInt();

            switch(choice)
            {
                case 1 :
                    System.out.println("\n1.Grade Evaluation System");

                    int maths = 80, science = 85, history = 90;
```

```
int avg = (maths + science + history) / 3;

System.out.println("Average Marks are : " + avg);

if(avg >= 90)

    System.out.println("Grade : A");
else if(avg >= 70 && avg <= 89)

    System.out.println("Grade : B");
else if(avg >= 50 && avg <= 69)

    System.out.println("Grade : C");
else if(avg >= 30 && avg <= 49)

    System.out.println("Grade : D");
else

    System.out.println("Fail");

break;
```

case 2 :

```
System.out.println("\n2. Leap Year Check");

int year = 1900;

if( year % 400 == 0)

{

    System.out.println(year + " : is a Leap Year");

}

else if(year % 4 == 0 && year % 100 != 0)

{

    System.out.println(year + " : is a Leap Year");

} else {
```



```

        System.out.println(year + " : is Not a Leap
Year");
    }
    break;

    case 3 :
        System.out.println("\n3.Day of the week ");
        int day = 3;
        switch(day)
        {
            case 1:
                System.out.println("The day is
Monday");
                break;

            case 2:
                System.out.println("The day is
Tuesday");
                break;

            case 3:
                System.out.println("The day is
Wednesday");
                break;

            case 4:
                System.out.println("The day is
Thursday");
                break;

            case 5:

```

```

        System.out.println("The day is Friday");
        break;

        case 6:
            System.out.println("The day is
Saturday");
            break;

        case 7:
            System.out.println("The day is Sunday");
            break;

        default :
            System.out.println("Invalid day
number");
            break;

    }
    break;

    case 4 :
        System.out.println("\n4. Identify Default Values of
Variables");

        byte a = 14;
        short b = 11;
        int c = 50000;
        long d = 123456789L;
        float e = 3.14f;
        double f = 60.5;
        char g = 'S';

```

```
boolean h = true;
```

```
System.out.println("byte: " + a);
```

```
System.out.println("short: " + b);
```

```
System.out.println("int: " + c);
```

```
System.out.println("long: " + d);
```

```
System.out.println("float: " + e);
```

```
System.out.println("double: " + f);
```

```
System.out.println("char: " + g);
```

```
System.out.println("boolean: " + h);
```

```
break;
```

```
default:
```

```
System.out.println("Invalid choice");
```

```
}
```

```
}
```

```
while (choice != 5);
```

```
{
```

```
System.out.println("\nThank You ...!");
```

```
}
```

```
}
```

```
}
```

Output:

```
C:\WINDOWS\system32\cmd.  ×  +  ∨  
Menu-Driven are :  
1.Grade Evaluation System  
2.Leap Year Check  
3.Day of the week  
4.Identify Default Values of Variables  
5.Exit  
  
Enter Your choice :  
1  
  
1.Grade Evaluation System  
Average Marks are : 85  
Grade : B  
  
Menu-Driven are :  
1.Grade Evaluation System  
2.Leap Year Check  
3.Day of the week  
4.Identify Default Values of Variables  
5.Exit  
  
Enter Your choice :  
5  
Invalid choice  
  
Thank You ...!  
  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```